

PE-EC701A — Microwave Theory and Technique

Semester VII • 3L:0T:0P • 3 Credits

Module 1 Exam Notes: Introduction to Microwaves

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Introduction to Microwaves

Microwaves represent a key segment of the electromagnetic spectrum that lies between radio frequencies and infrared radiation. They are characterized by short wavelengths, which gives rise to their name.

Microwaves: Electromagnetic waves with frequencies ranging from **300 MHz (0.3 GHz) to 300 GHz**, which corresponds to free-space wavelengths ranging from **1 m to 1 mm** respectively. They propagate in line-of-sight paths and are governed by Maxwell's electromagnetic equations.

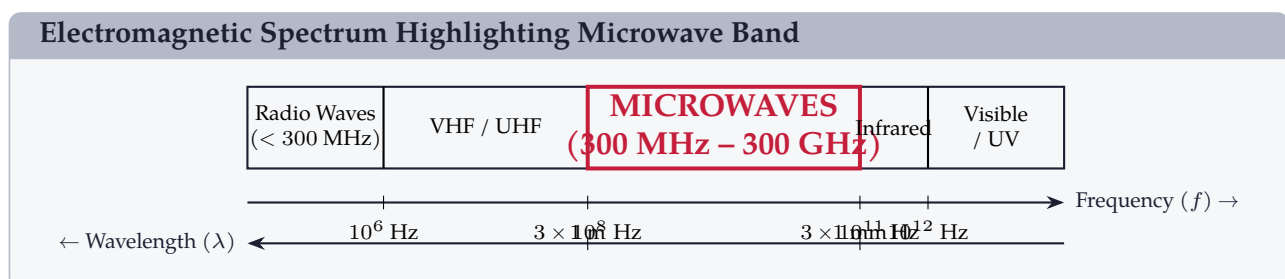
1.1 History and Evolution of Microwaves

The physical foundation of microwaves dates back to the theoretical and experimental milestones of the 19th and 20th centuries:

- **James Clerk Maxwell (1864):** Formulated the electromagnetic theory predicting the existence of electromagnetic waves.
- **Heinrich Hertz (1888):** Experimentally proved Maxwell's equations by generating and detecting radio waves at UHF/microwave frequencies using spark-gap transmitters.
- **Jagadish Chandra Bose (1895):** Generated millimeter-wave signals up to 60 GHz and demonstrated remote signaling and polarization effects using horn antennas and semiconductor detectors.
- **World War II (1939–1945):** Accelerated microwave research due to the operational necessity of Radar. The invention of the Cavity Magnetron by Randall and Boot enabled high-power radar transmitters.
- **Post-War Era:** Shifted focus toward civilian telecommunications, satellite links, cellular networks, and domestic heating (microwave oven).

1.2 Electromagnetic Spectrum Representation

The position of the microwave band within the wider electromagnetic spectrum is illustrated below:



2. Microwave Frequency Bands

To avoid spectral confusion, the microwave spectrum is subdivided into distinct bands. The two primary naming standards are the **IEEE Standard** (used in radar and communications) and the **Military/NATO Standard**.

2.1 IEEE Microwave Frequency Band Classification

The table below details the frequency and wavelength ranges defined by the IEEE:

Band Designation	Frequency Range	Wavelength Range	Typical Applications
L Band	1.0 – 2.0 GHz	30.0 – 15.0 cm	GPS, Mobile phones (GSM), Satellite navigation
S Band	2.0 – 4.0 GHz	15.0 – 7.5 cm	Wi-Fi (2.4 GHz), Microwave ovens, Weather radars
C Band	4.0 – 8.0 GHz	7.5 – 3.75 cm	Satellite TV transponders, Wi-Fi (5.8 GHz)
X Band	8.0 – 12.0 GHz	3.75 – 2.50 cm	Military radar, Marine navigation, Speed detectors
Ku Band	12.0 – 18.0 GHz	2.50 – 1.67 cm	Direct-to-Home (DTH) TV, Satellite networks
K Band	18.0 – 27.0 GHz	1.67 – 1.11 cm	Radar altimeters, Water vapour absorption line (22 GHz)
Ka Band	27.0 – 40.0 GHz	1.11 – 0.75 cm	High-speed satellite broadband, Close-range radar
V Band	40.0 – 75.0 GHz	7.5 – 4.0 mm	Short-range military comms, 60 GHz Wi-Fi (IEEE 802.11ad)
W Band	75.0 – 110 GHz	4.0 – 2.7 mm	Automotive radars, Millimeter-wave imaging

3. Key Advantages and Properties of Microwaves

Microwaves possess specific properties that differentiate them from lower-frequency radio waves:

1. High Information Carrying Capacity (Large Bandwidth): The available bandwidth in

the microwave region is several orders of magnitude larger than the entire HF and VHF radio spectrum combined.

$$\text{Bandwidth } \Delta f = f_2 - f_1 \quad (1)$$

Since frequency is high, a small fractional bandwidth yields enormous absolute bandwidth, enabling high-rate data transmission.

2. **Line-of-Sight (LoS) Propagation:** Microwaves travel in straight lines and are not reflected by the ionosphere. This ensures clean, interference-free point-to-point communication but requires repeater towers due to earth's curvature.
3. **Directive Antenna Gain:** The gain of an antenna is inversely proportional to the square of the wavelength:

$$G = \frac{4\pi A_e}{\lambda^2} \quad (2)$$

Because λ is small, very high antenna gain and narrow beamwidths can be achieved with physically small structures (e.g., dish antennas).

4. **Lower Power Requirements:** Highly directive antennas concentrate energy into a narrow beam, requiring significantly lower transmitter power to span long links compared to omnidirectional HF systems.
5. **Ionospheric Transparency:** Frequencies above 100 MHz pass directly through the ionosphere without being reflected or bent, making space, satellite, and deep-space communications possible.

4. Applications of Microwaves

4.1 Civil and Military Applications

The primary applications of microwaves are split across the civilian and defense sectors:

Civilian Applications:

- **Telecommunications:** Cellular networks (LTE, 5G), Wi-Fi router bands, and satellite voice/data networks.
- **Satellite Broadcasting:** Direct-to-Home (DTH) TV services (operating in the Ku band) and satellite radio.
- **Meteorology:** Weather radar systems operating in the S and C bands to map precipitation and storm tracks.
- **Domestic Heating:** Microwave ovens operate at 2.45 GHz, utilizing dielectric heating of water molecules in food.

Military Applications:

- **Radar Detection and Tracking:** Fire-control radars, air traffic monitoring, and target tracking (operating in X and Ku bands).
- **Electronic Warfare (EW):** Jamming enemy communications, electronic countermeasures (ECM), and radar signature reduction (stealth).
- **Missile Guidance:** Semi-active and active radar seekers located in the nose cones of missiles.

4.2 Medical Applications

Microwaves interact with biological tissues in ways that enable both diagnostics and therapy:

- **Microwave Hyperthermia:** Delivering controlled heat to tumors. Cancer cells are more heat-sensitive than healthy tissue, and raising their temperature to 42°C–45°C helps destroy them.
- **Microwave Ablation (MWA):** Using localized needle-like microwave antennas inserted into tumors to cook and destroy cancer cells directly.
- **Microwave Imaging:** Utilizing dielectric contrast between healthy and malignant breast tissues to detect tumors without ionizing radiation.

5. EMI/EMC Concepts at Microwave Frequencies

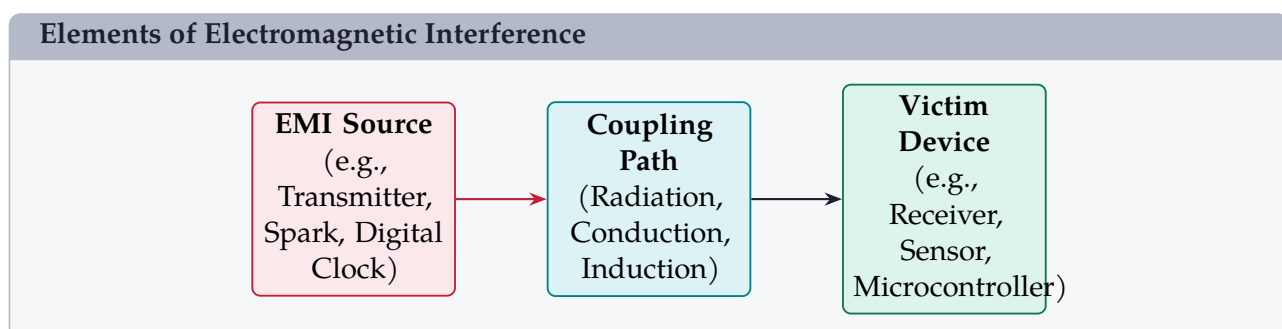
As operating frequencies increase, electromagnetic interactions become critical, necessitating strict EMI/EMC compliance.

EMI (Electromagnetic Interference): Any electromagnetic disturbance or emission that degrades, disrupts, or limits the effective performance of electrical equipment.

EMC (Electromagnetic Compatibility): The ability of an electrical system or device to function satisfactorily in its electromagnetic environment without introducing intolerable electromagnetic disturbances to anything in that environment.

5.1 The EMI Coupling Path

For electromagnetic interference to occur, three elements must be present:



5.2 EMI Mitigation Techniques

- **Shielding:** Enclosing sensitive circuits in metallic containers (Faraday cages) to reflect and absorb radiated fields.
- **Grounding:** Ensuring low-impedance ground return paths to prevent common-mode voltage differences.
- **Filtering:** Installing low-pass filters on power and signal lines to suppress high-frequency noise.

Quick Revision — Module I

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Key Formulas at a Glance

Wave Velocity Relation	$c = f\lambda \approx 3 \times 10^8 \text{ m/s}$ (in free space)
Antenna Directivity/Gain	$G = \frac{4\pi A_e}{\lambda^2}$ (A_e = effective aperture area)
Bandwidth	$\Delta f = f_2 - f_1$
Decibel Power Ratio	$P_{\text{dB}} = 10 \log_{10} \left(\frac{P_{\text{out}}}{P_{\text{in}}} \right)$

One-Line Definitions

- **Microwave:** Electromagnetic waves operating between 300 MHz and 300 GHz.
- **Dominant Mode:** The waveguide transmission mode that exhibits the lowest cutoff frequency.
- **Line-of-Sight:** Straight-line propagation between antennas with no ionospheric reflection.
- **EMI:** Disturbance of electronic systems caused by unintended electromagnetic fields.
- **EMC:** The capacity of a device to operate without radiating or absorbing excessive EMI.
- **Microwave Hyperthermia:** Thermal therapy utilizing microwaves to selectively heat and destroy tumor cells.
- **Faraday Cage:** A conductive enclosure used to block external radiated electromagnetic fields.

Frequently Asked / Viva Questions

Q1. State the frequency and wavelength ranges of microwaves.

Frequencies range from 300 MHz to 300 GHz, corresponding to free-space wavelengths of 1 m to 1 mm respectively.

Q2. Why are microwaves not reflected by the ionosphere?

Because of their high frequencies ($> 100 \text{ MHz}$). The ionospheric plasma frequency is lower, allowing microwaves to pass directly through without reflection.

Q3. What are the typical frequency ranges for the L, S, C, and X bands?

L band: 1 – 2 GHz; S band: 2 – 4 GHz; C band: 4 – 8 GHz; X band: 8 – 12 GHz.

Q4. Why are microwave antennas physically small for a given high gain?

Frequently Asked / Viva Questions (contd.)

Gain is inversely proportional to λ^2 ($G = 4\pi A_e/\lambda^2$). Since λ is small, a small aperture A_e is sufficient to achieve high gain.

Q5. List three major advantages of using microwaves for communication.

Large operational bandwidth (high information capacity), high antenna directivity, and immunity to ionospheric disturbances.

Q6. What is the operating frequency of a domestic microwave oven and why?

2.45 GHz. This frequency matches the rotational excitation energy of water molecules, ensuring efficient dielectric heating of food.

Q7. Explain the difference between EMI and EMC.

EMI is the interference signal itself that disrupts electronic operation. EMC is the design property of a system to resist EMI and avoid emitting it.

Q8. State the medical applications of microwaves.

Microwave hyperthermia for cancer therapy, microwave ablation to destroy tissue, and non-invasive microwave breast imaging.

Q9. Why is point-to-point microwave communication limited by distance?

Because microwaves propagate in straight lines (LoS). Due to the earth's curvature, repeater stations are required approximately every 50 km.

Q10. What is the significance of the X-band in military applications?

X-band (8–12 GHz) offers a good compromise between atmospheric attenuation and spatial resolution, making it ideal for tracking and fire-control radars.

Q11. How does high operating frequency affect digital circuits regarding EMI?

As frequency increases, trace lines on PCBs act as antennas, leading to crosstalk, radiation emission, and high susceptibility to noise.

Q12. What is shielding effectiveness (SE)?

It is the ratio of the electromagnetic field strength before shielding to the field strength after shielding, measured in decibels (dB).

Comparison of Civil and Military Microwave Applications

Parameter	Civil Applications	Military Applications
Primary Goal	High-capacity data transfer, media broadcast, and domestic heating.	Surveillance, threat tracking, weapons guidance, and communications jamming.
Typical Bands	L, S, C, Ku bands (high-speed data, DTH satellite).	X, Ku, Ka, W bands (fine-resolution radar, stealth systems).
Design Priority	Cost efficiency, standardized channels, and regulatory compliance.	High reliability, anti-jamming capabilities, and high-power output.
Standard Devices	Solid-state transceivers, standard Wi-Fi routers.	High-power magnetrons, traveling-wave tubes (TWT), and phased-array systems.